NEHA MIRIAM JOSE

Computer Science Engineer

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SUMMARY

Enthusiastic and detail-oriented B.Tech Computer Science student with a strong interest in problem-solving, programming, full-stack development, machine learning, deep learning, and cybersecurity. Hands-on experience through internships and academic projects involving NLP, audio deepfake detection, secure web applications, and LoRa-based IoT sensing. Strong academic record with a CGPA of 9.29/10.00. Eager to apply knowledge in real-world projects, learn new technologies, and contribute to impactful software solutions in a dynamic environment.

WORK EXPERIENCE

Data Science & Machine Learning Intern | NeST Digital, Kochi (Feb 2026 – Mar 2026)

- Preprocessed and analysed real-world datasets using Python and applied machine-learning algorithms to build and evaluate predictive models.
- Applied data preprocessing and model evaluation techniques for machine-learning tasks.

Machine Learning Intern | Infosys Springboard (Virtual) (Oct 2025 – Dec 2025)

- Developed an NLP-based intelligent ticket classification pipeline using SpaCy, NLTK, BERT, Logistic Regression, SVM, Random Forest, and XGBoost for multi-label categorization and priority prediction.
- Built an end-to-end web application using Streamlit (Frontend), FastAPI (Backend), and SQLite for real-time ticket processing, entity extraction, and dashboard analytics.
- Implemented data processing with scikit-learn, pandas, and imbalanced-learn (SMOTE) for class balancing, feature engineering, and model-pipeline optimization.

PROJECTS

Audio Deepfake Detection using Ensemble Deep Learning | *Python, TensorFlow/Keras, Librosa, NumPy, scikit-learn*

- Developing an ECN-MF-inspired audio deepfake detection system using MFCC, Chroma, and Zero-Crossing Rate (ZCR) features.
- Implementing and comparing RNN, 1D CNN, LSTM, and ConvLSTM models and combining predictions through soft-voting ensemble classification.
- Working with the Fake-or-Real audio dataset and evaluating models using accuracy, precision, recall, F1-score, AUC, confusion matrices, and ROC analysis.
- Conducting feature-combination/ablation experiments to study the contribution of MFCC, Chroma, and ZCR to detection performance.

Smart RF-Based Landslide Early Warning System | *ESP32, SX1278 LoRa, MPU6050, DHT22, Soil Moisture, Rain Sensor*

- Designing a LoRa-based early-warning system for hilly regions using ESP32 sensor nodes and a 433 MHz SX1278 wireless communication link.
- Integrating MPU6050 tilt sensing, soil-moisture, rainfall, temperature, and humidity measurements to monitor conditions associated with landslide risk.
- Developing the base-station concept using an ESP32 LoRa receiver, buzzer alert, and laptop dashboard for receiving and displaying sensor data.

AI-Powered Ticket Classification System | *Python, SpaCy, NLTK, BERT, FastAPI, Streamlit, SQLite, scikit-learn, Logistic Regression*

- Developed an intelligent ticket management application with automated categorization, priority prediction, and named entity recognition using multiple machine-learning models.
- Architected a FastAPI backend with SQLite for ticket data processing and model inference and integrated a Streamlit frontend for real-time analytics and user interaction.

Ethical Hacking Simulation Portal | *HTML, CSS, JavaScript, FastAPI, SQLite, Docker*

- Developed an ethical hacking simulation portal with 9+ interactive challenges covering real-world vulnerabilities including SQL Injection, XSS, CSRF, and command injection.
- Implemented a FastAPI backend with SQLite for user authentication, challenge progression, quiz assessments, tutorials, analytics, leaderboard rankings, and role-based admin controls.

Food Ordering System | *HTML, CSS, JavaScript, Flask, SQLite*

- Developed a comprehensive food ordering application with restaurant discovery, menu browsing, cart/order workflows, ratings and reviews, and order-management functionality.
- Built a Flask backend with SQLite for storing users, food items, orders, and related application data, with a responsive frontend for customer interactions.

EDUCATION

B.Tech – Computer Science and Engineering *2023 – 2027*
Amal Jyothi College of Engineering (Autonomous), Kanjirappally | **CGPA: 9.29 / 10.00**

Higher Secondary (Class XII – CBSE) *2023*
St. Antony's Public School Anakkal, Kanjirappally | **86.6%**

Secondary (Class X – CBSE) *2021*
St. Antony's Public School Anakkal, Kanjirappally | **95.4%**

SKILLS

Programming: Python, C, Java (Basic)
Machine Learning / Data: Machine Learning, Deep Learning, TensorFlow/Keras, scikit-learn, pandas, NumPy, Librosa
Web: HTML, CSS, JavaScript
Frameworks: Flask, FastAPI, Streamlit
Database: SQL, SQLite
Embedded / IoT: ESP32, SX1278 LoRa, MPU6050, DHT22, Sensor Interfacing
Tools: GitHub, VS Code, Figma, Docker, Google Colab
Concepts: Data Structures, DBMS, Web Vulnerabilities
Soft Skills: Teamwork, Adaptability, Time Management, Quick Learning

CERTIFICATIONS

- Cyfusion – UI/UX Design Competition (SAHYA'26) | Designed interfaces using Figma | 2026
- Database Management Systems – Elite | NPTEL (IIT Kharagpur) | 2025
- HTML5, CSS3 & JavaScript | Infosys Springboard | 2025
- Problem Solving through Programming in C | NPTEL (IIT Kharagpur) | 2024
- Career Essentials in Generative AI | Microsoft and LinkedIn | 2024

REFERENCES

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